

- 1(a) An embedded system is often a small device that is designed to carry out a limited number of specialised tasks. Professional athletes sometimes wear small embedded systems called fitness trackers in their shirts. These can be used to track their speed, position, heart rate and other performance data during an event. The tracker transmits this data to a pitch-side server which collates the data from all the athletes.

State the name of **one** input device that might be used in this embedded system and state what it would be used for.

Device

.....

Use

.....

[2]

- (b) The fitness tracker manufacturers had to decide which type of secondary storage to use.

Explain **two** reasons why flash (solid state) storage would be the best type of secondary storage for the fitness tracker system.

1

.....

.....

.....

2

.....

.....

.....

[4]

(c) One use of ROM is to store the Basic Input Output System (BIOS). The BIOS is used when the computer is first turned on.

i. Describe what the BIOS will do to start up the computer.

[3]

ii. Apart from storing the BIOS, ROM can also be used in other ways.

Describe how an embedded system can make a different use of ROM and why it is an advantage.

[2]

2(a) State the role of RAM in a computer system.

[1]

(b) Amaya is an amateur photographer and has bought an old second-hand computer to edit her photographs. The specifications of this computer are shown below.

Processor: Dual Core 1.8 GHz
RAM: 1 GB
HDD: 500 GB

Explain what is meant by the term ‘virtual memory’ and why this may be needed when Amaya is editing her photographs.

[2]

3 A computer system will contain several input and output devices.

Explain the role of device drivers when using input and output devices on a computer system.

[2]

4 A delivery company sends parcels across the UK.

To prove parcels have not been damaged in transit, the delivery drivers use a digital camera to take a photograph of them when they arrive at their destination. The digital camera uses flash memory.

i. Describe one advantage of the digital camera using flash storage rather than magnetic.

[2]

ii. Explain whether lossless or lossy compression would be most appropriate to store the photographs. Justify your response.

[3]

5 Before a video is rendered, the user will first capture and edit the individual video elements before they are combined together to form the final video.

i. State **two** different output devices that could be used when editing the videos.

1 _____

2 _____
_____ [2]

ii. A storage device is used to store the individual video elements while they are being captured in different locations and during the video editing process.

Discuss the suitability of a flash storage device **and** a magnetic storage device for storing the different video elements while they are being collected and edited.

You should refer to the following in your answer:

- the benefits of each type of storage
- the drawbacks of each type of storage
- the suitability of each type of storage.

[9]

- 6 * OCR Insurance's computer system uses secondary storage across the company in servers, client machines
and for backup purposes.

For each of these, discuss whether magnetic storage or solid state storage would be most suitable, taking into account the advantages and disadvantages of both.

This image shows a blank sheet of white paper with ten horizontal dashed lines, typical of primary-ruled notebook paper. The lines are evenly spaced and extend across the width of the page. There is no handwriting or other markings on the paper.

[9]

7(a) Traditionally films have been distributed on optical media such as DVDs.

Giving an example other than DVDs, describe what is meant by the term 'optical media'.

[2]

(b) Give **one** advantage of films being distributed using optical media.

[1]

(c) Explain the advantages and disadvantages of owning films that are streamed or downloaded on demand rather than owning a physical copy.

[4]

8 A company releases an in-home virtual assistant called 'Bertie Butler'.

The device, when placed in a room, listens out for the phrase "Hey Bertie". When someone says that phrase it then listens to the question that follows and tries to give a relevant answer.

Bertie Butler has a number of built-in input and output devices.

Name **one** input device and **one** output device that might be part of Bertie Butler. For each device give a reason for it being built into the virtual assistant.

Input Device Name: -----

Input Device Reason: -----

Output Device Name: -----

Output Device Reason: -----

[4]

9(a) A company produces digital photo frames (i.e. photo frames that display digital photographs).

Give **two** reasons why this operating system may be stored in ROM.

1

2

[2]

- (b) The photo frames can also play video clips. Due to the limited storage on the device the videos are stored on the company's servers (i.e. 'in the cloud') and streamed when needed.

Give **one** disadvantage to the user of using cloud storage for their photos.

----- [1]

- (c) The company allows users to connect a local storage device to the photo frame to increase storage capacity.

State the name of **one** device that might be connected to the photo frame to increase storage capacity.

----- [1]

END OF QUESTION PAPER

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
1	a	1 mark max for input device and 1 mark max for reason: • Heart rate sensor • Detects athlete's electrical activity / heart rate • GPS (receiver) • Mapping the athlete's movement / position/ speed • Accelerometer/gyroscope/motion sensor • To track the players acceleration/speed/movement/ counting steps/measuring orientation • Button • To allow the athlete to turn it on/off	2	Allow other suitable input devices as long as they are suitable for the scenario and an input device e.g. thermometers, light and UV sensors, skin response sensors, magnetometers, gyrometer, ECG No FT for use if incorrect device Use must be related to the fitness tracker scenario <u>Examiner's Comments</u> The majority of candidates were able to state the name of a device, but many did not apply the use of their named input device to the scenario and gave responses related to a generic use of the device for example giving a heart rate sensor or monitor and then saying the use was to monitor the heart rate with no reference to the fitness tracker. A few gave devices which were unsuitable for an embedded system carried in a shirt such as keyboard and touchscreen.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
	b		1 mark for reason and 1 mark for expansion up to a maximum of 4 marks. • Durable/flash has no moving parts... • ...so will be resistant to damage if the player runs/falls/collides with something • Low power usage... • ...won't need recharging/will last during an event • Small physical size/portable... • ...so it can fit on a shirt/be unobtrusive/be worn // won't affect the athlete's performance • Fast/real-time read/write speed... • ...needed to rapidly record athlete's data	4	Mark across whole answer space Expansion must relate to scenario DNA 'small' on its own DNA anything to do with capacity Examiner's Comments The majority of candidates were able to give a reason such as durable/no moving parts/ small physical size. Candidates who related the reason to why it would be suitable for the fitness tracker were able to gain full marks. Some candidates did not apply their response to the question stem well and instead talked about things like smart watches. Exemplar 2 1. Flash storage has no moving parts so it is more shock resistant than other storage devices (which either require moving components to read or write data). This means that it would be less likely to get damaged during activities like running that the others can get into. 2. Flash storage is also very small in size (but still at the same capacity) so the small device could be smaller. This would make it easier to carry (i.e., it is more portable due to smaller weight and dimensions) and therefore attract/retain more users. The candidate has clearly given two reasons with good application to the scenario.
	c	i	1 mark per bullet to max 3: • Contains the computer start up instructions • Loads settings/configuration (CMOS/NVRAM) • Initialises/checks hardware/peripheral devices are available/work // carry out a POST check... • ...and reports errors • Determines the drive on which the OS is stored • Finds/loads the bootstrap/operating system/kernel (into main memory)	3	DNA 'boots up' on its own Examiner's Comments Many candidates were able to gain at least one or two marks for saying the BIOS finds/loads the operating system and/or checks hardware.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
		ii	Mark from one group to max 2: - Storing firmware/ config/ operation instructions/ operating system/ device drivers... ...that can't be overwritten // doesn't need to be updated ...so that access is faster OR - Storing security settings/encryption keys/digital signatures... ...that must be maintained/can't be changed OR - Could be used as <u>primary</u> storage... ...so no need to load programs into memory ...so would save time loading up the program OR - Could be used as <u>secondary</u> storage... so the device will not need additional storage... to reduce the size of the device	2	<u>Examiner's Comments</u> Strong candidates were able to give a use such as store the operating system and an advantage. Many though, gave a very good description of what ROM is but did not give a different use and so were unable to gain any marks. A few candidates said it could store the BIOS which had already been given in the question. A few candidates talked about microwaves and washing machines which was not relevant to this scenario
			Total	11	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
2	a		- Stores instructions / programs currently running - Stores data currently in use	AO1.1 (1)	<u>Examiner's Comments</u> This question was answered well, although some candidates stated the role of cache rather than RAM.  Misconception That RAM and cache are the same thing within a computer system.
	b		- If the amount of RAM available is insufficient... ...Used to store instructions/data ...Using secondary storage. - Likely to be loading large image files (which may exceed her RAM) - Likely to be loading large programs (which may exceed her RAM)	AO2.1 (2)	<u>Examiner's Comments</u> Most candidates achieved both of the available marks for this question and linked the scenario in well when writing their response.
			Total	3	
3			- A piece of software which allows hardware/device to communicate... ...with the operating system	AO1.2 (2)	<u>Examiner's Comments</u> This question was generally answered well. Some candidates were not given the second mark as they did not make reference to the operating system.
			Total	2	

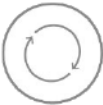
Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
4		i	– Has no moving parts... (1) ... less likely to be damaged / data loss (1) – Consumes less power... (1) ...meaning battery needs recharging less often (1) – Unaffected by magnetic fields.. (1) – ...which could lead to data loss. (1) – Faster access speed... (1) – ...so photographs can quickly be written to it / browsed. (1) (Mark in pairs)	2 AO1.2 (1) AO2.1 (1)	<u>Examiner's Comments</u> Many candidates gave a reason for the advantage without stating the actual advantage e.g. 'flash storage has no moving parts' without going on to say, therefore the advantage is 'less likely to be damaged/lose data'.
		ii	– Lossy compression (1) Any 2 of the following, must be in context: – Some loss of detail is acceptable (1) – Unlikely to be noticeable (1) – Will make the file size smaller than lossless(1)	3 AO1.2 (1) AO2.1 (2)	Do not accept 'smaller' on its own, must be 'smaller than lossless'. Do not accept lossless compression as most appropriate. In this case give zero marks for the justification also. <u>Examiner's Comments</u> Candidates would be best advised to fully consider how they would justify their choice appropriately for the given scenario before committing to 'lossy or lossless'. Those who correctly chose 'lossy' went on to achieve at least one of the other two available marks.
			Total	5	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
5		i	• Monitor • Speakers//Headphones	AO2.1 (2)	Must be output devices. Must be suitable for video editing. <u>Examiner's Comments</u> Many candidates answered this question correctly, with most selecting a monitor/screen and headphones/speakers which were both valid responses.
		ii	Mark Band 3–High Level (7–9 marks) The candidate will provide a range of accurate benefits and drawbacks of both magnetic and flash devices. The candidate is able to apply their knowledge and understanding directly and consistently to the context provided. Evidence/examples will be explicitly relevant to the explanation. The candidate will come to a clear conclusion that must be justified by their comments <i>There is a well-developed line of reasoning which is clear and logically structured. The information presented is relevant and substantiated.</i> Mark Band 2–Mid Level (4–6 marks) The candidate will provide drawbacks or benefits about both magnetic and flash methods. These will be for the most part relevant. The candidate is able to apply their knowledge and understanding directly to the context provided although one or two opportunities are missed. Evidence/examples are for the most part implicitly relevant to the explanation. The candidate will attempt to come to a conclusion, although it may not be fully justified by their answer. <i>There is a line of reasoning presented with some structure. The information presented is in the most part relevant and supported by some evidence.</i> Mark Band 1–Low Level (1–3 marks) The candidate has provided some drawbacks or benefits of magnetic or flash methods. The material is basic and	AO1.1 (2) AO1.2 (2) AO2.1 (2) AO3.3 (3)	AO1 Magnetic Benefits: • Lower price per GB • As standard comes in high capacity Magnetic Drawbacks: • Lower R/W speed • Higher power consumption • Moving parts /less durable Flash Benefits: • Higher R/W speed • No moving parts/more durable • Lower power consumption • Can have a higher capacity. Flash Drawbacks: • Higher cost per GB • At higher capacities prices rise exponentially. AO2: Suitability of Magnetic: • Video files are very large, benefiting magnetics low cost per GB • Can be used to archive the different video elements after it's been rendered • Could possibly get damaged while being transported between the different filming sites • Slow R/W speeds may take it longer to load into the software Suitability of flash: • Very fast R/W speeds therefore reducing the amount of loading time during the editing process/giving

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
			contains some inaccuracies. The candidate makes a limited attempt to apply acquired knowledge and understanding to the context provided. The candidate provides nothing more than an unsupported assertion. <i>The information is basic and communicated in an unstructured way. The information is supported by limited evidence and the relationship to the evidence may not be clear.</i> 0 mark No attempt to answer the question or response is not worthy of credit.		smoother playback • If money is not a barrier, flash can be bought in sufficient capacities for videos. AO3: Candidates can conclude either method, but to score in the top MB must have a clear line of reasoning to justify their choice. <u>Examiner's Comments</u> Candidates were assessed on the quality of their extended response in this question. Most candidates understood the basic differences between flash storage and magnetic storage devices, but did not make clear links to the scenario. Mid-level responses were common, with few candidates giving a balanced discussion and a conclusion with suitable justifications.  Assessment for learning Questions with the command word 'discuss' require candidates to give a balanced discussion and provided a suitable conclusion which justifies their comments. Opportunities to practise these questions will support candidates to do better on these style questions.
			Total	11	

Mark Scheme

Question	Answer/Indicative content	Marks	Guidance
6	Mark Band 3–High Level (7-9 marks) The candidate demonstrates a thorough knowledge and understanding of magnetic and solid state storage. The material is generally accurate and detailed. The candidate is able to apply their knowledge and understanding directly and consistently to the context provided. Evidence/examples will be explicitly relevant to the explanation. The candidate provides a thorough discussion which is well balanced. Evaluative comments are consistently relevant and well-considered. There is a well-developed line of reasoning which is clear and logically structured. The information presented is relevant and substantiated. Mark Band 2-Mid Level (4-6 marks) The candidate demonstrates reasonable knowledge and understanding of magnetic and solid state storage; the material is generally accurate but at times underdeveloped. The candidate is able to apply their knowledge and understanding directly to the context provided although one or two opportunities are missed. Evidence/examples are for the most part implicitly relevant to the explanation. The candidate provides a sound discussion, the majority of which is focused. Evaluative comments are for the most part appropriate, although one or two opportunities for development are missed. There is a line of reasoning presented with some structure. The information presented is in the most part relevant and supported by some evidence. Mark Band 1-Low Level (1-3 marks) The candidate demonstrates a basic knowledge of magnetic and solid state storage; the material is basic and contains some inaccuracies. The candidate makes a limited attempt to apply acquired knowledge and understanding to the context provided.	AO1.1 (2), AO1.2 (2), AO2.1 (2), AO3.3. (3)	Data stored by altering the magnetic field of a small section of the tape or disk; one orientation represents a binary value of 1, the opposite orientation represents a binary value of 0. Written to using a read head. Read head can then read the magnetic field and retrieve the data. These read and write heads move across the disk or the tape moves under the heads. Solid state devices store data using semiconductors; drives can be based on NAND Flash or DRAM. No moving parts are required. Backup involves copying potentially large amounts of data to and from storage devices at regular intervals. Servers distribute data to client devices. AO2 Magnetic hard drives entirely suitable for storing operating system, software and data. Large capacity, cheap (by comparison) purchasing price and reliability make them idea for heavy use in an insurance office. Solid state drives have no moving parts and offer an increase in speed and quicker access to customer records. However, smaller capacity (in comparison) may be an issue for larger data sets. Backup systems may be required to maximise storage whereas access speed is not so much of an issue Limited lifespan of SSDs due to limited read/writes may restrict use in data heavy applications. Servers will require quick access to data to serve multiple client machines but will also be extremely read/write heavy. AO3 Both suitable for the insurance office but for different purposes. SSDs to store operating system and program files will ensure quicker boot up and loading of programs. May be useful for both client and servers to store operating systems Magnetic hard drives to store large quantities of data provides reliability and large capacity at small (by comparison) cost. May be most suitable for storing large amounts of data on client devices

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
			The candidate provides a limited discussion which is narrow in focus. Judgments if made are weak and unsubstantiated. The information is basic and communicated in an unstructured way. The information is supported by limited evidence and the relationship to the evidence may not be clear. 0 marks No attempt to answer the question or response is not worthy of credit.		and for backup systems. Magnetic tape device may be suitable for backing up customer data as too slow for use on day-to-day basis. Use of solid state devices for storing customer data that is frequently accessed means needing to plan for replacement of drives when read/write limit reached. Solid state devices better if used in mobile devices (e.g. for mobile insurance workers) as relatively small chance of damage if dropped as no moving parts.
			Total	9	

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
7	a	One from – read / written to, using a laser – read / written onto a reflective surface – data is stored using pits and lands Plus example – Eg. Blu-Ray / CD (1 per - , max 2)	2 AO1.1	Accept legacy media such as laserdisc and videodisc. Do not accept DVD variants (DVD-R, DVD-RW etc)
	b	– Cheap to produce / buy – Capacity big enough to hold film – Robust enough to be used over long periods / moved around – Portable – Does not need an internet connection (1 per - , max 1)	1 AO1.1	<u>Examiner's Comments</u> Many candidates achieved full marks on both part (a) and (b) demonstrating their understanding of optical media.
	c	Advantages – Can access film (and indeed entire library) from anywhere with an internet connection – No physical storage needed – No risk of films being stolen / damaged/lost – Optical / disk player hardware not required to play film Disadvantage – Reliant on having an (fast enough) internet connection. – Need enough storage for download – Reliant on servers of the company providing the connection. – People like having a physical copy – If the company providing the stream goes out of business, copy is lost. (1 per -, max 4. If only advantages or disadvantages, max 3)	4 AO1.2	<u>Examiner's Comments</u> Most candidates achieved at least one advantage and one disadvantage mark. Most commonly citing access to the internet renders the film(s) accessible from anywhere. Although some candidates referred to 'on line' access which is not creditworthy.
		Total	7	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
8			Input Device Microphone... ...To allow the device to hear spoken requests. Buttons... ...To turn the device off or on / To mute device / to put device in privacy mode (max 1 device name, max 1 reason) Output Device Speaker... ...To play the device's responses LEDs... ...To indicate the device is on/listening Screen... ...To show visual information (max 1 device name, max 1 reason)	4 AO1.2 (2) AO2.1 (2)	Do not award "...listen out for the phrase "Hey Bertie"" Award any appropriate input / output device <u>Examiner's Comments</u> Virtually all candidates correctly named appropriate input and output devices. Fewer candidates went on to give appropriate reasons, many were too generic to be creditworthy at this level of study e.g. 'Speaker – to output the response'
			Total	4	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
9	a		- The software will not need updating - Can be loaded quickly when the frame is switched on/will not be lost when the frame is switched off. - Cannot be (accidentally) overwritten/deleted <i>(1 per -, max 2)</i>	2 AO1.2 (2)	<u>Examiner's Comments</u> Many candidates scored at least one mark here, with some repeating the same point. Candidates should be reminded that, where appropriate, their response should be in context of the scenario. Those that were not lost credit.
	b		- Potentially vulnerable to hacking. - Reliant on an internet connection/requires a fast enough internet connection. - Reliant on company servers. <i>(1 per -, max 1)</i>	1 AO2.1 (1)	<u>Examiner's Comments</u> Most candidates achieved the single mark available in this question.
	c		- (External) hard (disk) drive/ solid state drive - Flash memory card/stick/drive <i>(1 per -, max 1)</i>	1 AO1.2 (1)	Accept SD Card/Micro SD Card Do not accept just 'USB', must be USB stick/device//pen/drive <u>Examiner's Comments</u> Most candidates achieved the single mark available in this question. Although some candidates incorrectly offered simply 'USB' as a response. USB is an interface, not a device.
			Total	4	